



EIRION



AI-POWERED PERSONALIZED HEALTH SPAN OPTIMIZATION

TEAM NAME: CRYPTARCHS

EVENT: ET GEN AI HACKATHON

PROBLEM STATEMENT

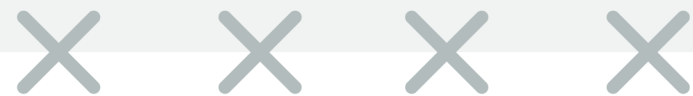
- Current healthcare gap (no predictive tools for long-term toxicity)
- Financial waste (\$5K-50K annually without guidance)
- Genetic variation ignored (50% metabolize differently)
- One-size-fits-all prescriptions leading to adverse reactions

Who it impacts?

- Biohackers & longevity seekers (taking 10-20 supplements daily)
- Chronic medication users (133M in US, 500M+ globally)
- Athletes optimizing performance
- Aging population seeking healthspan extension

REAL-WORLD APPLICATION - CONCRETE USE CASE

- User Profile: 35-year-old professional, takes Omega-3 (2000mg), Vitamin D (5000 IU), Ashwagandha (600mg), Metformin (off-label 500mg), high-sugar diet (150g/day), moderate exercise
- Current State: Feels healthy, normal blood work, but unaware of cumulative risks
- AI Analysis: CYP2D6 poor metabolizer detected, Ashwagandha accumulation risk, sugar-medication interaction
- Prediction: "Liver efficiency drops 15% by age 40, elevated AST/ALT by age 43"
- Personalized Recommendations:
 - Switch Ashwagandha to Rhodiola (safer for genotype)
 - Reduce sugar to 90g/day (30% reduction)
- **Outcome:** Biological age stays 3 years younger than chronological through optimized interventions, liver function maintained at 98% efficiency through age 50



PROPOSED SOLUTION

EIRION (Proposed GenAI Approach – Multi-Modal AI System Architecture)

Datasets & Sources:

- PharmGKB: 750+ drug-gene associations
- CPIC Guidelines: Clinical pharmacogenetics
- ToxCast/Tox21: 10K+ toxicity assays (EPA) ; FDA FAERS: 10M+ adverse event reports
- User Data: Genetics (23andMe), wearables (Fitbit), lab results

Experiments & Validation:

Phase 1: Train models on 50K+ patient records (85%+ accuracy target)

Phase 2: 6-month clinical pilot (100 users) - track liver/kidney function, biological age vs AI predictions

Evaluation Metrics:

- Prediction accuracy: 85%+ (toxicity events)
- 30% reduction in adverse reactions vs control
- Biological age improvement (0.5-1 year)
- MAE <10% for organ function forecasts

Novelty - What Makes This Unique:

- First Predictive Health Simulator - forecasts "what happens in 5 years", not just current state
- Temporal Modeling - aging trajectories vs snapshot assessments
- Prevention-Focused - optimizes healthspan (massive market vs disease treatment)

Scope to Scale:

Year 1-2: Foundation

- B2C platform for biohackers
- 50 drugs, 3 organ systems
- 1,000 subscribers → \$350K ARR

Year 3-5: Scale

- 200+ drugs, full multi-omics integration
- 100K+ subscribers globally

Impact: Democratize personalized longevity medicine, prevent 1M+ adverse reactions, extend healthspan by 5+ years for 100K+ users